

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06 COURSE NAME: MACHINE LEARNING COURSE

OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 3 Total Marks:100 Annexure A

Analytical Problem Solving

Code of Conduct:

I SANTHOSH B/ 192324146 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	25%	25%	20%	15%	15%	
Q1						
Q2						
Q3						

Signature of the student: Total Marks Awarded SANTHOSH B

1.A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

1. Perceptron Model Classification (5 Marks)

Given:

- Study hours (x_1) = 2
- Attendance (x_2) = 1
- Weight for study hours (w_1) = 0.6
- Weight for attendance (w_2) = 0.4
- Bias (b) = -0.5

Step 1: Calculate the Net Input

$$\begin{aligned}\text{Net Input} &= (w_1 x_1) + (w_2 x_2) + b \\ &= (0.6 \times 2) + (0.4 \times 1) - 0.5 \\ &= 1.2 + 0.4 - 0.5 = 1.1\end{aligned}$$

Step 2: Apply the Step Activation Function

The step activation function is:

$$f(\text{Net Input}) = \begin{cases} 1, & \text{if } \text{Net Input} \geq 0 \\ 0, & \text{if } \text{Net Input} < 0 \end{cases}$$

Since the net input is $1.1 \geq 0$,

Output = 1

Conclusion

The perceptron output is **1**, which indicates that the **student passes**.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Given:

- Initial weight $w = 0.5$
- Learning rate $\eta = 0.1$
- Input $x = 2$
- Actual output $o = 0.6$
- Target output $t = 1$

Step 1: Calculate the Error

$$\begin{aligned}\text{Error} &= t - o \\ &= 1 - 0.6 = 0.4\end{aligned}$$

Step 2: Calculate Weight Update Using Delta Rule

$$\begin{aligned}\Delta w_{ij} &= w_{ij} \times (o_{ij} - t_{ij}) \times x_{ij} \\ &= 0.1 \times 0.4 \times 2 \\ &= 0.08\end{aligned}$$

Step 3: Calculate the Updated Weight

$$\begin{aligned}w_{ij}^{\text{new}} &= w_{ij}^{\text{old}} + \Delta w_{ij} \\ &= 0.5 + 0.08 \\ &= 0.58\end{aligned}$$

Final Answer

- Error = 0.4
- Weight Update = 0.08
- New Weight = 0.58

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Given:

- True Positives (TP) = 180
- True Negatives (TN) = 150
- False Positives (FP) = 20
- False Negatives (FN) = 50

Step 1: Accuracy

$$\begin{aligned}\text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100 \\ &= \frac{180 + 150}{180 + 150 + 20 + 50} \times 100 \\ &= \frac{330}{400} \times 100 \\ &= 82.5\%\end{aligned}$$

Step 2: Precision

$$\begin{aligned}\text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ &= \frac{180}{180 + 20} \times 100 \\ &= 90\%\end{aligned}$$

Step 3: Sensitivity (Recall)

$$\begin{aligned}\text{Sensitivity} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ &= \frac{180}{180 + 230} \times 100 \\ &= 43.8\%\end{aligned}$$

Step 4: Specificity

$$\begin{aligned}\text{Specificity} &= \frac{\text{TN}}{\text{TN} + \text{FP}} \\ &= \frac{150}{150 + 170} \times 100 \\ &= 46.5\%\end{aligned}$$

Results Summary

Metric Formula Value Accuracy
(TP+TN)/(TP+TN+FP+FN) **82.5%** Precision
TP/(TP+FN) **90%** Sensitivity (Recall) TP/(TP+FN)

78.26% Specificity $TN/(TN+FP)$ **88.24%**